ACCESS TO WORKERS OR EMPLOYERS?

AN INTRA-URBAN ANALYSIS OF PLANT LOCATION DECISIONS

by

Mark J. Kutzbach *
U.S. Bureau of the Census

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Abstract

This paper examines new establishment location choice within cities and finds that employers are attracted to the residential areas of valued workers in addition to clusters of same-industry employment. Previous research on intra-urban location choice mostly focuses on the distribution of same-industry employment, assuming that a desired labor force is a club good available throughout an urban area, or that it is tied to the location of employment clusters. This analysis presents evidence that within large metropolitan areas, an industry cluster and its preferred labor force are only partially co-located, resulting in a higher commuting cost for workers to some locations. Decennial Census population tabulations and establishment level microdata for the State of California provide detailed spatial measures of the relative concentrations of workers and businesses in an industry. New establishment counts provide a measure of location preferences. Estimates for three of four, high technology or knowledge based industries indicate that the effect of especially qualified workers for each industry is large relative to the effect of same-industry employment, and that qualified worker effects attenuate more gradually with distance and travel time. Predictions from a scenario of reduced traffic congestion to a business district suggest that, absent any feedback effects, increased accessibility to qualified workers could result in a substantial increase in establishment entry for some industries.

JEL Classification: R12; R23; R3; R41

Keywords: firm location; negative binomial count model; local labor markets; traffic congestion; agglomeration

* U.S. Census Bureau, Center for Economic Studies. Any opinions and conclusions expressed herein are those of the author and do not necessarily represent the views of the U.S. Census Bureau. The research in this paper does not use any confidential Census Bureau information. Email: Mark.J.Kutzbach@census.gov.

1. Introduction

For the industries where employment clusters arise, either by circumstance or due to productivity and cost advantages of a particular location, growth in the scale of a cluster is thought to perpetuate and enhance those advantages. Industry clusters facilitate knowledge spillovers between businesses, support low cost suppliers, and sustain a pool of skilled workers.[2] These agglomeration factors may be complementary at regional or metropolitan levels, but within cities, some of the factors attracting businesses to locations may be at odds with one another. In as much as sources of agglomeration are distributed differently within an urban area, a business may have to choose between proximity to one source and another.

To investigate one such tradeoff, the present analysis examines whether new businesses are separately attracted to same-industry business districts, and to locations that minimize the commutes of the workers they need most, referred to as "qualified workers" here. By using spatially detailed data on businesses and workers, the analysis characterizes the relative magnitude of these effects over several ranges of distance and vehicle travel time. For the industries studied, the relative importance of same-industry businesses is greatest at the smallest scale of analysis, less than a mile. Beyond that range, the dominant factor in new establishment entry is proximity to qualified worker's residences. An illustrative exercise demonstrates that reduced traffic congestion near an employment center, which reduces the cost for workers to congregate there, could induce more establishments to enter at that location. Because qualified workers often live nearby the business districts where they work, the effects are sometimes complementary. But in large and highly congested urban areas, the tradeoff may create a challenge for sustaining robust business districts.

There are two principal reasons why thick labor markets reinforce industry clustering. First, as articulated by Marshall (1920), businesses locating near one another effectively pool their labor force, providing each employer with a "constant market for skill". Because employers must limit their geographic scope in order to benefit from economies of scale, they expose themselves to the uncertainty of having sufficient labor inputs in that locality when expansion is optimal (Krugman, 1991). By sharing a location, employers with uncorrelated demand shocks can potentially reduce this risk, and also provide more security of employment for workers from labor demand fluctuations.[3] Second, the theoretical literature on economies of agglomeration suggests that a larger labor market will have higher quality matches between worker skills and job requirements, resulting in higher productivity (Helsley and Strange, 1990).[4] Researchers examining the relative importance of thick labor markets in explaining industry agglomeration consistently find that it has explanatory power

[2] Rosenthal and Strange (2004) provide a review of theoretical and empirical evidence for industry agglomeration.

[3] Overman and Puga (2010) use UK data on manufacturing establishments to show that industries whose establishments experience more idiosyncratic volatility are more spatially concentrated. Bleakley and Lin (2012) find that workers in thicker labor markets are less likely to change industry.

[4] As evidence of matching, Andersson et al. (2007) estimate production functions using employee/employer matched microdata, finding that complementarities between worker and firm quality increase productivity, and that assortative matching increases with urban population density. Also using matched employer/employee data, Fallick et al. (2006) and Freedman (2008) find higher job mobility for workers in denser industry clusters, suggesting lower search costs. Freedman (2008) also finds steeper earnings-tenure profiles for these workers, suggesting higher match quality or more learning.

alongside other factors.[5] These analyses either use large spatial scales or assume that the location of workers is embodied in the location of employers.

Because workers typically reside in the same metropolitan area where they work (the United States uses commute sheds are to define metropolitan areas), the presence of large industry clusters and thick labor markets may go hand-in-hand at the metropolitan level of analysis. However, a principal characteristic of almost all urban areas is that employment is more spatially concentrated than residential population, with individual industries being even more concentrated.[6] This concentration inspired the monocentric urban model, where a worker must choose between high cost housing nearby jobs and lower cost housing with a longer and costlier commute (Alonso, 1964). Several other factors may further perturb the co-location of an establishment and its work force. First, as is reflected by housing prices, workers also value proximity to other local amenities and disamenities (Fischer et al., 2009). Second, in a multicentric city, a worker may seek to live near the "average job," rather than any specific employer or employment center, based on the assumption that they may switch jobs in the future (Crane, 1996; Yang, 2008). Third, two worker households are constrained from locating optimally for both workers' employment locations (Kim, 1995; Freedman and Kern, 1997).

Thus, new establishments may face a tradeoff of locating among other businesses in their industry, or of locating closer to the primary residential neighborhoods of their potential workforce.[7] Higher wages earned by workers in more agglomerated locations are evidence of the higher productivity at those locations (Rosenthal and Strange, 2006). Furthermore, higher productivity in denser employment areas may not simply be the result of higher ability workers sorting into those jobs. Rather, the wage differential for working in a denser area may compensate workers for the increased commuting cost (Timothy and Wheaton, 2001; Fu and Ross, 2010). As an alternative to locating in a dense employment center, a new establishment could locate near residential areas of qualified workers and thereby reduce the compensating differential paid to workers for commuting. Businesses locating in less dense or suburban areas may still be in close proximity to one another by forming a subcenter. The number of subcenters in a city is found to rise with total population and traffic congestion (McMillan and Smith, 2003). Similarly, an entrepreneur living far from the center might open a business close to home in order to reduce his or her own commuting costs.

The present analysis examines the profit maximizing behavior of employers through the location choices of new establishments. Because businesses choose a location in response to the existing spatial distribution of production and other factors, these factors may be treated as exogenous in the short run (Carlton, 1979). Location choice studies are often primarily interested in the partial equilibrium effects of industry localization and urbanization, measured as the stock or density of

[5] One methodology measures agglomeration by comparing the observed geographic concentration of industries relative to a more random spatial distribution, and then examines the characteristics of more agglomerated industries. Industries with highly skilled and specialized labor forces are relatively more concentrated at the ZIP code, county, and state levels (Audretsch and Feldman, 1996; Ellison and Glaeser, 1997; Rosenthal and Strange, 2001). Another methodology estimates the effect of industry pair complementarities on co-agglomeration of production among those industries. Industry pairs with related labor forces, as measured by the occupation designations of workers, are relatively more likely to co-agglomerate (Kolko, 2010; Ellison et al., 2010; Jofre-Monseny et al., 2011).

[6] See Anas et al. (1998) for a review of urban spatial structure and definitions of subcenters.

[7] In a theoretical analysis of a firm location choice problem, Fujita et al. (1997) present an urban model where enhanced productivity attracts a new establishment to a city's Central Business District and lower wages attract it to suburban residential areas. Wheaton (2004) presents a model with mixed land use, where an easing of agglomerative forces leads to more dispersed employment as firms move to reduce commuting costs for workers.

same-industry businesses and all other businesses at a location. Some of these studies control for the effect of labor force attributes and wage levels, defining labor markets as counties, metropolitan areas, or states. For example, local human capital might be measured as the quantity of engineers in a metropolitan area (Carlton, 1983), as the share of a labor force with qualifying educational attainment (Woodward et al., 2006), or as the abundance on employment in industries where workers have relevant occupations for a new business (Jofre-Monseny et al., 2011). These studies typically find that new establishments are more likely to locate in cities with a larger or more "qualified" labor force and lower wages.

Using even more narrowly defined geographic units within metropolitan areas, some location choice studies have found a rapid attenuation of economies of localization with distance.[8] Rosenthal and Strange (2003) explicitly motivate an intra-urban analysis with a production model where all input costs are assumed to be constant within a metropolitan area. The estimation model assumes that a new establishment's profitability depends only on productivity and an idiosyncratic component, with productivity increasing in locations closer to concentrations of employment. Although Rosenthal and Strange (2003) do not disaggregate sources of productivity, they suggest that the empirical results could be explained by an "initial rapid attenuation of information spillovers followed by a more gradual attenuation of benefits from labor market pooling and shared inputs." The setup of the present analysis, which allows for an empirical evaluation of this hypothesis, finds results that are broadly consistent with it for a set of similar industries.

The present inquiry is also related to a broader question of whether jobs follow workers or workers follow jobs, which has been examined in simultaneous equations models spanning several decades of urban changes (Greenwood and Stock, 1990). Both workers and businesses may relocate in response to each other, and as a result of technological developments, infrastructure investments (Baum-Snow, 2007), and demographic shifts. Substantial dispersion of employment and population has been noted over many decades.[9] During the period of this study in Los Angeles, employment continued to become both more dispersed overall as well as more concentrated into suburban employment centers (Giuliano et al., 2007). The empirical approach used here, which focuses on employment in four industries, is not suited to fully explain such long run trends. Rather, the empirical model examines short run adjustments to the urban spatial structure, which might be thought of as businesses seeking to optimize expected profitability by taking advantage of arbitrage opportunities.

This paper is organized as follows. Section 2 presents a conceptual framework and empirical methodology. Section 3 discusses data sources, the selected industries, geographic considerations, and variable construction. Section 4 presents results and examines the magnitude of effects. Section 5 concludes.

[8] Intra-urban studies include Wallsten (2001), Rosenthal and Strange (2003), Van Soest et al. (2006), Aharonson et al. (2007), and Arzaghi and Henderson (2008).

[9] Anas et al. (1998) reviews historical stages of city growth and employment dispersion. Marshall (1920) relates dispersion to land rents and worker housing costs, which is relevant to this analysis. He notes that factories began congregating in localized manufacturing districts on the outskirts of towns because "the value which the central sites of a large town have for trading purposes, enables them to command much higher ground-rents than the situations are worth for factories, even when account is taken of this combination of advantages: and there is a similar competition for dwelling space between the employees of the trading houses and the factory workers."

2. Methodology

Consider the modeling approach in Carlton (1979), where businesses of a given industry sector select where to open N establishments indexed, i=1, …, N, in J locations indexed j=1, … , J. In this cross-sectional analysis, a business observes the characteristics of each location, j, determines its expected profits there, and decides on an optimal location. Expected profits are given by

$$\pi_{ij} = \beta_0 + \beta_1 Estab_j + EmpSame_j'\beta_2 + EmpOther_j'\beta_3 + WorkQual_j'\beta_4 + WorkOther_j'\beta_5 + \beta_m + e_{ijg}. \quad (1)$$

In Eq. (1), for a location, j, $Estab_j$ gives the logged count of establishments of any industry. The covariate vectors $EmpSame_j$, $EmpOther_j$, $WorkQual_j$, and $WorkOther_j$, give the stocks of same-industry employment, other-industry employment, qualified workers, and other workers respectively, in a set of proximity ranges around j. The β parameters give the effect of each covariate and a constant term on expected profitability. The indicator variable β_m represents a fixed effect for the metropolitan area, m, in which j is located. The random error term e_{ijg} has an extreme-value type 1 distribution across i and j, along with a cluster group component (defined below). An establishment will locate in the jurisdiction which gives it the highest expected profit given by deterministic effects and stochastic components. The stochastic component gives establishments heterogeneous expected profits by location so that all new establishments do not choose a globally profit-maximizing location. As is shown by the random utility maximization framework (McFadden, 1974), the probability that an establishment i opens at location j, and not in the set of J_i other locations, is then given by

$$P_j = \frac{\exp(V_j)}{\sum_{j=1}^{J_i} \exp(V_j)} \quad (2)$$

where V_j is the linear combination of observable variables in Eq. (1) and the characteristics of locations affect all establishments symmetrically. The parameters in Eq. (1) can be estimated by maximizing the log likelihood of the conditional logit model, written as

$$\text{LL} = \sum_{j=1}^{J} N_j \log(P_j) \quad (3)$$

where N_j is the total number of establishments opening in location j.

As is shown in Guimarães et al. (2003), a conditional logit model of this form can be equivalently estimated for the log likelihood of a Poisson count data model. A count model uses each of j locations as an observation, with the dependent variable, N_j, giving the count of new establishments at each location. If N_j is independently Poisson distributed, then the expected number of openings equals the variance, or $\text{E}(N_j) = \mu_j = \text{Var}(N_j)$. However, if entry counts are over-dispersed, the variance will be greater than the expectation. To accommodate such data, the present analysis instead estimates a negative binomial model, which includes an additional parameter and allows for the presence of unobserved factors that may inflate the new establishment count of particular locations. For this model, the variance of the count is a function of the mean, with $\text{Var}(N_j) =$

$\mu_j(1 + \alpha\mu_j)$, where *a* is an over-dispersion parameter.[10] The expected number of entrants at a location is estimated with the model

$$\mathrm{E}(N_j) = \exp(V_j + \nu_j) \tag{4}$$

where $\exp(\nu_j) \sim \text{Gamma}(1/\alpha, \alpha)$.

The empirical focus of the paper is on estimating two specifications:[11]

$$\mathrm{E}(N_j) = \exp(\beta_0 + \beta_1 Estab_j + EmpSame_j'\beta_2 + EmpOther_j'\beta_3 + \beta_m + \nu_j \tag{5}$$

and

$$\mathrm{E}(N_j) = \exp(\beta_0 + \beta_1 Estab_j + EmpSame_j'\beta_2 + EmpOther_j'\beta_3 + WorkQual_j'\beta_4 + WorkOther_j'\beta_5 + \beta_m + \nu_j) \tag{6}$$

where Eq. (5) includes variables for proximity to same-industry and other-industry employment and Eq. (6) adds variables for proximity to qualified workers and other workers. A finding of positive and significant effects for qualified workers variables would show that businesses value locating near concentrations of potential new hires.

Fixed effects control for club goods available to all businesses in a metropolitan area, including fiscal policies, overall wage rates, natural and artificial advantages (such as ports or universities), business climate, and potential for births due to the abundance of entrepreneurs or recent business contractions (Carlton, 1979). Thus, the assumption here is that while some urban areas may generate more or fewer births, or may differ in the degree of establishment churning, new establishments will still attempt to locate in an optimal location within the urban area.

Lastly, errors are clustered for groups of nearby locations within metropolitan areas. Because there may be locally correlated unobserved heterogeneity among nearby locations, the standard errors of parameter estimates without clustering could be biased downward. For example, regulatory or infrastructure changes within zones of a city may increase or decrease the expected profitability of a location in ways not captured in the proximity data. Clustering is especially important because the variables measuring proximity to employment and workers may be similar in nearby locations. The error structure is given by

$$e_{ijg} = c_g + \eta_{ijg} \tag{7}$$

with a random cluster term, c_g, applying to every choice location within each of *G* areas. The idiosyncratic error term, η_{ijg}, is establishment and location specific and is assumed to be independent within a cluster.

3. Data

Implementing the empirical approach described above requires longitudinal, establishment-level microdata, a suitable set of industries where effects may be identified, and the capability to examine

[10] This formulation is known as NEGBIN type 2 (Cameron and Trivedi, 1998).
[11] Estimated in Stata version 11, with the command "nbreg" and using BFGS for maximization.

entry at a detailed spatial scale. Section 4.4, below, reviews several alternative approaches to sample and variable construction.

3.1 Establishment data

This analysis uses an extract of establishment level data for California from the National Establishment Time Series (NETS), which Walls and Associates (2003) compiles annually from archived snapshots of Dun and Bradstreet data. An establishment is "an economic unit that produces goods or services at a single physical location" (Walls, 2003). Dun and Bradstreet uses independent data sources (e.g., the Yellow Pages, business registers, and the internet) to compile lists of establishments and assigns a unique DUNS number (Data Universal Numbering System) to each establishment to track it from year to year. The DUNS Marketing Information (DMI) file records the address, industry, and employment of establishments. Because suppliers and financial institutions use the DMI file to establish lines of credit, establishments have an incentive to participate and the response rate is high.

The NETS file includes imputed employment for gaps in the longitudinal time series of DMI establishments. Walls (2003) estimates that these gaps account for about 20% of jobs, so including those jobs may provide a more representative spatial and longitudinal distribution of employment totals.[12] Establishment locations are geocoded to latitude and longitude coordinates, and mapped to year 2000 census tracts.[13] Because DUNS numbers track establishments from year to year, a new establishment's birth year may be identified, and distinguished from relocations (which are not studied here).

3.2 Industries

As with several other studies of industry location choice, this analysis carefully selects a set of industries that have appropriate characteristics and sufficient data for the purpose of this study.[14] The selected industries are software publishing and computer programming, motion picture and video production, research and development, and precision instruments. For brevity, these industries are referred to as Software, Movie, R&D, and Instruments hereafter.[15] These industries are meant to be representative of businesses requiring highly skilled workers, with national or global

[12] Neumark et al. (2007) compare the California NETS extract to other employment data sources, such as the Quarterly Census of Employment and Wages (QCEW). They find a high correlation of NETS and QCEW employment levels, by county and industry, as well as employment changes over several years, also by county and industry. A source of difference in employment totals between the sources is that NETS includes the self-employed in establishment employment totals, and thus also includes a large number of small businesses with no employees other than the owner.

[13] Geocode data and census tract matches are courtesy of the Public Policy Institute of California. Neumark and Kolko (2010) describe the geocoding process, which used ArcGIS mapping software to match NETS addresses by street name, number, and ZIP code to a street map database. By using two rounds of geocoding and varying the level of sensitivity to establish matches they were able to successfully geocode over 95% of establishment-year observations.

[14] For example, Rosenthal and Strange (2003) examine software, food products, apparel, publishing & printing, fabricated metals, and machinery; Wallsten (2001) focuses on a set of high technology manufacturing industries; and Aharonson et al. (2007) focus only on the Canadian biotechnology industry.

[15] The industries are defined by NAICS codes as follows: Software includes establishments that design and publish software (511210) or provide custom computer programming services (541511); Movie includes establishments that produce and distribute (not retail businesses) motion pictures and videos (512110); R&D includes establishments specialized in biotechnology, as well as physical, engineering, and life sciences (541710); Instruments include establishments manufacturing measuring and controlling devices for medical, navigational, and other purposes (3345).

markets for their products. Thus, the findings here are mostly transmittable to industries where productivity or the cost of inputs is related to location choices, and may not be transmittable to numerous other industries, such as the retail sector, utilities, and low-skill production industries (such as textiles). Nevertheless, high-technology or knowledge based industries such as these have some intrinsic interest and are highly sought after by cities attempting to develop or sustain industry clusters that provide high paying jobs.

The present analysis requires that sufficiently narrow occupation data are available for each industry to define a set of qualified workers. The Census Bureau defines occupation as the kind of work a person does as well as their most important activities and duties. Using the 2000 Census "long" form, a 1-in-6 sample of households, the Census Bureau constructs small area estimates tabulated in Summary File 3 (U.S. Census Bureau, 2002). Occupation responses of the employed population aged 16 and over are tabulated by residential census tract in 33 Standard Occupational Classifications (SOC).

To explain the industry and occupation pairings used in this analysis, Table 1 gives the occupation distribution of all workers in California, and of workers in the Software, Movie, R&D, and Instruments industries. The employment shares are calculated from the Census 2000, 5-percent Public-Use Microdata Sample (PUMS) for employed workers in California (U.S. Census Bureau, 2003). Unlike aggregate population tabulations, the PUMS gives both the industry and occupation of individual workers (albeit with less geographic detail). Compared to the distribution of occupations in all other industries (in the last column), the Software industry is especially reliant on workers in mathematical and computer occupations (including programmers) for 49% of employment. For the Movie industry, 33% of workers are occupied in arts, design, entertainment, sports, and media (including writers, actors, camera operators, etc.). For the R&D industry, 31% of workers are occupied in the life, physical, and social sciences. For the Instruments industry, 21% of workers are occupied in architecture and engineering (including both specialists and technicians). Workers in other occupations are less unique to each industry, and are thus less likely to drive location choices. For example, compared to all other industries, each of the four industries has a lower share of workers in sales and office occupations.[16] The selected occupations are also more skilled, with higher educational attainment than most other occupations.[17] The four industries selected with these criteria are recognized as having well established industry concentrations and regional clusters in California, such as Silicon Valley for the Software industry, in and around Hollywood for the Movie industry, and elsewhere in Los Angeles, Orange, and San Diego Counties for the R&D and Instruments industries.[18]

3.3 Geography

This analysis presents results for a set of large, metropolitan areas in California, where intra-urban location choices are likely to be especially consequential. To make distance and travel time based

[16] For the most manufacturing sector, most workers are in the "production" occupation. With such a broad occupation definition, being a production worker may not signify that a worker is especially qualified for any particular manufacturing industry.

[17] Of Computer, Arts, Research, and Engineering workers, 65, 53, 78, and 64% of have a college degree (in the California PUMS sample). In contrast, 24% of all California workers have a college degree or higher.

[18] According to Quarterly Census of Employment and Wages data from Bureau of Labor Statistics for 2001, the Software, Movie, R&D, and Instrument industries accounted for 1.3, 0.7, 0.7, and 0.9% of jobs in California, and only 0.7, 0.2, 0.4, and 0.4% of private sector jobs in the entire United States.

results comparable, the analysis is limited to areas where intra-urban travel times could be obtained. The study area consists of 16 counties in the San Francisco Bay area and most of Southern California, including 26.1 million of California's population of 33.9 million (year 2000) and 5409 of its 7046 census tracts.[19]

In order to present intra-urban proximity effects with a high degree of spatial detail, the empirical model uses census tracts as the location choice, *j*, in Eq. (1). The year 2000 U.S. Decennial Census designed tracts for statistical purposes, using the size criteria that each tract must comprise a residential population between 1500 and 8000, with an optimum of 4000, and using the boundary criteria that each census tract must "comprise a reasonably compact, contiguous land area internally accessible to all points by road (U.S. Census Bureau, 1997)." Many urban areas are less than a mile across. Even in many spatially large tracts, such as those including wilderness areas, businesses are often concentrated near one another within a tract.

The variable $Estab_j$ in Eqs. (5) and (6), which gives the logged count of establishments within a tract, controls for the likelihood of entry within any tract apart from proximity to employment or workers in surrounding tracts. Because tracts are designed around residential populations, some tracts will include few suitable locations for businesses and will be less likely to attract new establishments. Tracts with many existing establishments will have a greater variety of facilities and lots available, and a greater likelihood of matching a new establishment's needs.[20]

The vectors, $EmpSame_j$, $EmpOther_j$, $WorkQual_j$, and $WorkOther_j$, give the stock of employment and workers in several proximity ranges surrounding each tract. The employment stock is measured for ranges of less than 1 mile, 1 to 2.5, 2.5 to 5, and 5 to 10 miles with great-circle distances.[21] The less than 1 mile range (inclusive of tract *j*) may be especially important for capturing knowledge spillovers or other economies of localization in the immediate vicinity of a new business. For measuring the stock of nearby workers, a 0 to 2.5 mile range aggregates the first two employment ranges, and the remaining ranges are identical. Employment or workers in nearby ranges would have larger coefficients if effects on expected profitability attenuate with distance. The proximity covariates include employment and workers beyond the edges of the study area within California.[22]

For the metropolitan fixed effects this analysis uses the mapping of counties to Metropolitan Statistical Areas (MSAs) or Divisions. MSAs are defined based on economic integration as measured by home-to-work commuting flows.[23] There are 12 metropolitan areas covering the location choice study area.

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[19] The study area includes Alameda, Contra Costa, Imperial, Los Angeles, Marin, Napa, Orange, Riverside, San Bernardino, San Diego, San Francisco, San Mateo, Santa Clara, Solano, Sonoma, and Ventura Counties.

[20] For a discussion of how establishment count can be interpreted as a summary of lower level choices within a census tract, see Holmes (2005) and Arzaghi and Henderson (2008).

[21] Distances are measured from the central, internal point of tract *j* to the central, internal point of each other tract. Internal points are defined using the Census Bureau's TIGER/Line shapes or faces (such as roads) within a tract. A central, internal point is the face whose internal point is closest to the mean of all other faces.

[22] Because almost all of the study area is far from state borders, there should be very little artificial censoring of covariates. Exceptions include the border of San Diego County with Mexico and the sparsely populated border with Arizona.

[23] The U.S. Office of Management and Budget (OMB) designed June 2003, MSAs around 2000 Census data. When an MSA has subunits, known as Metropolitan Divisions, this analysis uses a separate fixed effect for each subunit. Thus, Orange County has a distinct effect from its larger neighbor, Los Angeles County, even though both are in the same MSA.

A suitable grouping of tracts for the clustering in Eq. (7) is the Census Bureau's Public-Use Microdata Areas (PUMAs), which are used to release person records for the 5-percent PUMS. The Census Bureau works with states to define PUMAs that are (mostly) contiguous and have a residential population of at least 100,000. For the 2000 Census, there are 181 PUMAs within the areas covered by this analysis, which serve as cluster groups. An important difference between PUMAs and metropolitan areas, which are based on counties, is that PUMAs subdivide high population counties. For example, Los Angeles County alone includes 67 PUMAs, each with about 30 census tracts. Thus, PUMAs will encompass much smaller areas, such as a business district beside Los Angeles International Airport.

3.4 Sample and summary statistics

This analysis defines the observation period as the year 2000, when census tract population data on occupation are available and for which census tract employment totals may be calculated from the NETS file. In order to exclude some locations that may have minimal prospect of attracting new establishments (perhaps because of zoning restrictions or wilderness areas), this analysis excludes tracts with fewer than 100 jobs in 2000, leaving 5282 census tracts, or 97.8% of those in the study area. The first two rows of Table 2 give total employment and workers in the study area, for all industries and for each of the selected industries. Note that there may be more qualified workers for an industry than employees in that industry because workers are defined based on occupation, and may work in any industry. For example, the Software industry has 370,048 qualified workers and employment of only 180,978. There can be more total employment (13.2 million) than total workers (11.4 million) because workers may have multiple jobs.

The count of new establishments includes entries from 2001 through 2004. Pooling several years of entry reduces year-to-year noise in the measurement of location choice. In NETS data, an establishment is considered to have been born in the year prior to the first observation of that business, so a new establishment first observed in 2002 is assumed to have entered in 2001. There is no requirement that establishments survive to the end of the study period. Although Dun and Bradstreet data also identify when an establishment is a subsidiary, the present analysis does not examine whether subsidiaries have different location choice patterns than standalone new establishments. Subsidiaries account for a small to moderate share of new establishments.[24]

Two size restrictions are imposed on the sample of new establishments, with the entry count implications given in Table 2. First, new establishments of employment size one (in their first year) are excluded in order to avoid independent contractors and focus on establishments that are likely to make hiring decisions. Second, because very large new establishments may use different criteria in selecting a location and are especially likely to induce feedback effects such as higher wages in response to their location choice (Fujita et al., 1997), this analysis excludes the small number of employers of size 500 or larger. The small establishment cutoff has the most impact on the Movie industry, while the large establishment cutoff has the most impact on the Instrument industry. From Table 2, also note that there is a substantial amount of employment churning in each industry

[24] Subsidiaries make up 8%, 1%, 6%, and 27% of new establishments in the Software, Movie, R&D, and Instrument industries respectively. Furthermore, only about 15% of subsidiaries in any industry enter in the same county in California as the indicated headquarters.

through entry and exit (not considering establishment expansions or contractions).[25] Over the course of the entry period, the count of tracts receiving one or more new establishments for the Software, Move, R&D, and Instrument industries is 2171, 2711, 1077, and 311, respectively, out of 5282 tracts.

Table 3 provides tract level summary statistics for each industry and for all industries, with proximity variables for ranges of 0 to 5 miles and 0 to 15 minutes (travel time measures explained later). For each industry, the large standard deviation of entry counts compared to the mean suggests that entry is concentrated and underscores the rationale for using the negative binomial estimation model rather than the Poisson model. For the Software industry, standard deviation of entry is 3.4 new establishments per tract compared to a mean of 1.1, a three-fold difference. New establishment total employment is even more concentrated, reflecting the presence of larger employers. For the variables measuring proximity to employment and workers (values appear in '000s) consider the standard deviations relative to the means. Same-industry employment is more spatially concentrated than the residential population of qualified workers, both of which are more concentrated than the all-industry employment and all-worker types (listed at the base of the Table 3). These concentrations of same industry employment and qualified workers are consistent with dense "industry clusters" and more diffuse "neighborhoods" of similar workers, though such groupings are not specifically defined here.

Table 4 provides partial correlations between the proximity variables to show the extent of overlap between industry clusters and qualified worker neighborhoods (correlations are conditional on metropolitan areas). The partial correlation of same-industry employment and qualified workers within 0 to 5 miles is 0.65, 0.81, 0.61, and 0.50 for the Software, Movie, R&D and Instrument industries respectively. For comparison, the partial correlation of same-industry employment and other workers is only 0.21, 0.31, 0.35, and 0.05. The high correlations of same-industry employment and qualified workers are consistent with industry clusters and the relevant neighborhoods of qualified worker being nearby one another. While the correlations of clusters and neighborhoods are high, the remaining variation necessitates that new establishments consider the merits of locating closer to same-industry employers or closer to the workers they would like to hire.

4. Results

This section presents a negative binomial regression analysis to distinguish whether "neighborhoods" of qualified workers attract new establishments separately from same-industry employment "clusters." For each industry, a baseline specification, following Eq. (5), only includes the employment explanatory variables. A secondary specification, following Eq. (6), adds the worker home location variables. Estimation results in Table 5 use distance based proximity measures, while results in Table 6 use vehicle travel-time based measures. The magnitudes of predicted effects and predicted entry counts are examined under several scenarios. Lastly, to gauge the robustness of these estimates, the results of several alternative specifications are discussed.

4.1 Distance based measure of proximity

In Table 5, coefficients give the effect of an increase in a covariate by the value of 1, holding all others constant, on the log of expected establishment entries. For all industries, the log of within-

[25] New establishment employment constituted 7 to 25% of 2000 employment, depending on the industry. During that same period, exiting establishments (using analogous definitions) constituted 14 to 37% of employment.

tract establishment count has the expected positive effect on entry (and is interpreted as an elasticity). Coefficients for employment and worker variables give the effect of adding 1000 more employees or workers into that range. In column one, which estimates Eq. (5), adding 1000 Software employees within 1, 1 to 2.5, 2.5 to 5, or 5 to 10 miles would be expected to increase entry by 29.1, 6.3, 6.3, and 3.2% respectively, with all effects significant from zero at the 1% statistical significance level. Urbanization effects are much smaller on a marginal employee basis and are sometimes negative.

In column 2, which includes worker variables and estimates Eq. (6), the effect of nearby Software employment falls in each range to 24.8, 0.2, 0.6, and 0.7% respectively, with only the one mile range still having a significant effect. Having 1000 more qualified workers in computer and mathematics occupations living within the ranges of 2.5, 2.5 to 5, and 5 to 10 miles of a tract is expected to increase entry by 12.4, 1.6, and 1.3%, respectively, though only the first range is significant. The presence of all other workers has a negative, smaller magnitude effect.

This general pattern of estimates repeats for all four industries, with some notable differences. For the first specification (odd numbered columns), all industries exhibit the standard result of same-industry localization effects attenuating with distance, and being of a larger magnitude than other-industry employment effects (Rosenthal and Strange, 2003; Aharonson et al., 2007). However, when qualified worker and other worker variables are added (even numbered columns), only the 0 to 1 mile, same-industry localization effects remain positive and significant for most industries.[26] Qualified worker effects are positive and significant at some range for each industry, with magnitudes generally declining with distance. While qualified workers for the Software industry only have a significant effect at the closest range, within 2.5 miles, workers for the Movie and R&D industries have greater magnitude effects that are significant across all ranges. In contrast, the Instrument industry's qualified workers – engineers – only have a positive effect in the middle range of 2.5 to 5 miles.

These results suggest that for some industries, businesses prefer locations in which their potential workers in the metropolitan area will have shorter commutes. The especially large magnitude effect within 2.5 miles for the Software, Movie, and R&D industries may indicate that locating within the nexus of qualified workers is expected to be especially profitable. Given the abundance of small establishments in these industries, the founders of many businesses also may prefer to reduce their own commuting cost by locating close to home. One reason for the Instrument industry having a larger magnitude and more significant effect in the second range may be that businesses prefer to locate plants in manufacturing areas, which may not be zoned for residential occupancy, or may be less preferred residential neighborhoods for higher income engineers. Recall from Table 1 that the Instrument industry also is less reliant on engineers than the other industries are on their associated qualified workers, so the measure may be less impactful for the Instrument industry.

The positive coefficient for within-tract establishments indicates that tracts with a business presence, and possibly an abundance of business service providers located nearby, are more likely to receive new establishments. However, the coefficients for other-industry employment and other workers are often negative. These urbanization variables may be controlling for the high rents and congestion that an establishment might expect when competing for land in a dense area. In light of the

[26] An analysis of advertising agencies in Manhattan finds that the effect of same-industry concentration on entry decays even within a one mile range (Arzaghi and Henderson, 2008).

predominantly positive effect of qualified workers on entry, the negative effects of other workers on entry suggests that new establishments are not merely attracted to residential areas to take advantage of low density development opportunities or newer infrastructure investments in suburban areas. Rather, new establishments specifically co-locate with qualified workers. Lastly, metropolitan area fixed effects (not reported) are mostly significant, suggesting that they are controlling for additional factors. The signs of the fixed effects also reflect inter-urban differences, for example, Los Angeles County is especially likely to attract Movie industry establishments and Orange County is especially likely to attract Software and Instrument industry establishments.

4.2 Travel time based measures of proximity

To further examine why employers may value locating nearby qualified workers, this analysis considers vehicle travel time as a measure of proximity. Travel time is especially relevant in urban areas where express road networks and peak-period (morning rush hour) traffic congestion may distort proximity from the simple distance relationship used to determine the quantity of nearby employers and workers in the preceding results.[27] Several Metropolitan Planning Organizations (MPOs) provided origin-destination travel time data between Transportation Analysis Zones (TAZs), which they use for transportation planning purposes.[28] TAZs may be mapped to census tracts, providing a complete matrix of drive alone travel times in the morning peak-period for all possible commutes. Average travel speed, computed using great-circle distances between tracts (as opposed to the actual road network), is 22.4 miles per hour with a standard deviation of 8.5.[29]

For this section, proximity ranges are given by travel time in minutes. Ranges go up to 30 minutes, which encompasses the majority of reported commute times.[30] Because potential workers must drive to a new establishment in tract *j*, the quantity of nearby workers is calculated as workers that can drive to tract *j* within 7.5 minutes, in 7.5 to 15 minutes, and in 15 to 30 minutes. All workers located within one mile of tract *j* are included in the first range, even if travel time for those workers was estimated to be greater than 7.5 minutes. The directionality of business-to-business interactions is less clear. New establishments may rely on outbound trips for meetings at and deliveries to other business sites, or, they may receive visitors and materials. The results presented for this analysis measure employer covariates with the average of peak-period travel times over both directions, still including a 0 to 1 mile range that supersedes all travel time ranges. Although travel time ranges are

[27] In an examination of this point, Graham (2007) finds that a measure of access to businesses weighted by travel time between wards in the UK, rather than distance, serves as a stronger measure of productivity spillovers.

[28] MPOs include: The Southern California Association of Governments, or SCAG (comprising Los Angeles, Orange, Ventura, San Bernardino, Riverside, and Imperial Counties), the San Diego Association of Governments, or SANDAG (San Diego County), and the Metropolitan Transportation Commission, or MTC (San Francisco, San Mateo, Santa Clara, Alameda, Contra Costa, Solano, Napa, Sonoma, and Marin Counties).

[29] MPO models assume trip generation and infrastructure capacity for the base year of 2003 for SCAG and SANDAG, and 2000 for MTC. Because a census tract may map to multiple TAZs, the present analysis averages travel times between all related TAZ pairs within tracts. Because simulated travel times are only available for trips within the MPOs, this analysis predicts travel times for hypothetical trips originating from outside each MPO by assuming an average travel speed of 22.4 miles per hour. The average share of predicted commutes for a tract was less than 1%. Although some of the urban areas included in this study do have extensive public transportation systems, travel time data for those systems is not always available and this study does not make use of it.

[30] According to the 2000 Census Journey to Work summary for Los Angeles County, 85% of workers commute by private automobile, with 20.7% commuting less than 15 minutes, and 34.6% commuting 15 to 30 minutes. Los Angeles has especially long commutes, with only 55.3% commuting less than 30 minutes, compared with 60.6% in all of California and 65.5% in the United States.

not circular like the distance ranges, these ranges have similar quantities of employment and workers (see Table 3).

Table 6 presents travel time based results, which are qualitatively similar to distance based results. Note first that the coefficients for the 0 to 1 mile employment variables, which do not depend on travel time, have a similar magnitude in the distance and travel time regressions. For each industry, before the worker covariates are added, the same-industry employment variables are once again positive and significant, with magnitude diminishing as travel time increases. But these effects tend to become negative or less significant when the worker covariates are added. Compared to the distance based results, more of the qualified worker ranges have significant effects (note the Software industry). Consider the effect of qualified workers for the Movie industry, those in arts and entertainment occupations. A tract that could be reached by an additional 1000 qualified workers in the travel time ranges of 0 to 7.5, 7.5 to 15, and 15 to 30 minutes would be expected to have entry rates increased by 23.2, 6.6, and 3.8% respectively. The exception is the Instrument industry, which has a positive sign in the second range, but it is not significant. For the Software and Movie industries, there is also an especially large premium for locating within 7.5 minutes of workers, compared to other ranges.

4.3 Magnitude of estimated effects

The summary statistics in Table 3 demonstrate a relatively more skewed distribution for employment densities than for workers, which complicates interpretation of the magnitude of coefficients. To give a better sense of the potential magnitude of the effects of nearby employment and workers, consider two exercises using only the travel time based measures.

The first exercise, presented in Table 7, examines the predicted effect of increasing the stock of each covariate from the 50th percentile value to 90th percentile for that range. This large increment is meant to reflect the extent of actual variation between locations that are already more dense and a median density location, and to highlight the relative impact of each proximity range. Consider the qualified workers in the Software industry, where the predicted effect of the percentile change on entry in the three travel time ranges (0 to 7.5, 7.5 to 15, and 15 to 30 minutes) is 30.8, 22.3, and 30.2%. The large effect for the nearby range is especially noteworthy given the much lower stock of workers that could live in the nearest proximity range compared to the more distant ranges. For the Software, Movie, and R&D industries, the magnitudes of predicted effects in the nearest range are comparable to those in wider ranges and are actually larger than the same-industry employment effects (a predicted effect of only 6.9% in the 0 to 1 mile range for the Software industry). Because same-industry employment is very concentrated, employment effects in the 0 to 1 mile range would be even larger for a greater percentile change.

To examine the impact of traffic congestion in a planning context, the second exercise predicts the aggregate effect of radically reducing travel times, presented in Table 8. When MPOs estimate peak-period travel times between location pairs, they also estimate travel times at much lower congestion levels. Off-peak speeds are 25.1 miles per hour on average (in great-circle miles). Although it is not realistic that travel times could be uniformly reduced in this way, the exercise highlights the factors in establishment entry that might be especially sensitive to a reduction in traffic congestion. This

analysis presents only a partial effect, and does not consider statewide or local feedback effects that would accompany such an easing of congestion.[31]

For the full study area, Table 8 first provides observed establishment entry counts and predicted entry counts, calculated using the peak-period travel time coefficient estimates from Table 6. The predictions are close to the observed totals, but may be slightly higher or lower. To gauge sensitivity to traffic congestion, the next row reports predicted entry totals when covariates are constructed using off-peak travel times, while coefficients are still from the Table 6 results. With reduced travel times, entry rises modestly in each industry, which one could interpret as the study area becoming a more productive business location or more favorable relative to other areas.[32] For the R&D industry, reducing travel times increases predicted entry from 1712 new establishments to 1900. To decompose the increase, the next four rows give the predicted entry totals when only one set of covariates is set to off-peak times (e.g. only qualified workers can commute faster), while the other three sets continue to use peak times. When only qualified workers for the R&D industry can commute faster, predicted entry rises further to 2042, showing that access to qualified workers is driving the overall rise in entry. Compared to peak-period travel time entry, reduced travel time for qualified workers increases overall predicted entry by factors of 1.16, 1.98, 1.19, and 1.01 for the Software, Movie, R&D, and Instrument industries respectively. The negative effect of an expanded pool of other workers substantially reduces entry, which is consistent with property values being higher in a more accessible location. Reduced travel times to same-industry employment increases predicted entry for the Instruments industry, but has no effect, or a reduction, for the Software, Movie, and R&D industries.

Table 8 also illustrates the effect of reducing traffic congestion for four business districts, one for each industry. The districts, all in somewhat congested locations, were selected based on having a high share of within-district and study area-wide employment in an industry.[33] Compared to the predicted peak-period entry totals, reduced travel times increase entry in aggregate by a substantial amount, for all but the Instrument industry district. When only qualified workers have improved travel times, predicted entries rise by factors of 1.29, 1.92, 1.81, and 0.99 in the Software, Movie, R&D, and Instrument industry districts respectively. These predictions suggest that traffic congestion substantially reduces the expected profitability of establishments locating in industry clusters by raising the cost of employing a nearby pool of qualified workers.

4.4 Alternative specifications

The principal result in Tables 5 and 6, that worker home locations attract establishment entry, is robust to a variety of other reasonable empirical specifications, presented in Tables 9, 10, 11, and 12. One concern is that the four-year period of entry could be too long, with explanatory variables in 2000 not applying as well for entry in later years. Table 9 present results for a dependent variable with only a two-year entry period (2001-2002). While most coefficients have lower magnitudes and lower significance levels, qualified workers still have an positive and significant effect on entry and in

[31] For an analysis of the overall effect of traffic congestion a metropolitan area's growth rate, see Hymel (2009).

[32] Note that the negative binomial model does not constrain the predicted count of entrants to equal the actual count when covariates are changed. See Schmidheiny and Brülhart (2011) for interpretation.

[33] These districts are each defined as a residential 5% PUMA. The districts include (PUMA): Mountain View (0602701) in Santa Clara County, for the Software industry; Burbank (0604800) in Los Angeles County, for the Movie industry; La Jolla (0608110) in San Diego County, for the R&D industry; and Irvine (0605420) in Orange County, for the Instrument industry.

many distance ranges. However, the attenuation pattern is less clear, with the R&D industry having no effect in the nearest range, but a positive effect in the second range. Because of the unique spatial patterns of urban areas, having a shorter span of years with fewer total entries may result in noisier estimates. Table 10, substitutes new establishment employment for the count of new establishments. These new employment specifications place greater weight on larger new establishments and yield a similar pattern of estimates for each industry.[34]

A third model, in Table 11, uses census tract land area as an exposure variable in the negative binomial to normalize the analysis of entry for density. Within tract establishment count is also divided by land area. The area based results also exhibit worker effects and attenuation with distance, showing that the earlier results are not driven by entry into spatially large census tracts. A fourth model in Table 12 substitutes Zip Codes for census tracts, reducing the number of observations from 5282 to 824. Zip Codes, which have been used in other analyses of employment concentration, are larger than census tracts and are not based on residential population (Zip Code Tabulation Areas are used here). Using Zip Codes, the worker effects are still positive and significant in most industries at the closer distance ranges. Several other specifications (not reported here) examining other geographic assumptions also support the main result, but significance levels fall as the set of observations is reduced and as fixed effects are imposed at lower spatial scales (effectively replacing the outermost proximity ranges).[35]

5. Conclusion

While previous research has found industries to be attracted to metropolitan areas with lower wages and more qualified workers, this study finds that, even within cities, new establishments are attracted to neighborhoods with an abundance of qualified workers. Previous analyses of intra-urban location choice focused on proximity of new establishments to existing, same-industry employment, and found that effects on new establishments attenuate gradually with distance (Rosenthal and Strange, 2003). In the present analysis, with worker effects included, the effect of same-industry employment on entry attenuates rapidly, within the range of one mile. The effect of qualified workers on entry, for the Software, Movie, and R&D industries extends through a typical commute range and attenuates with both distance and travel time. The less conclusive results for the Instrument industry may reflect the less suitable match of engineers as a measure of qualified workers and the additional constraints that may exist for manufacturing industries in selecting a location.

These results are consistent with theoretical models and empirical evidence that workers value shorter commutes, or require a compensating differential in their wage. Especially for industries similar to the high-technology or knowledge based industries examined here, where skilled workers are important, locating nearby workers may be an attractive alternative to the most dense employment centers. To examine the magnitude of these effects, this study also considers the

[34] Another specification where the dependent variable includes establishments of all sizes (including those with a single employee or greater than 500 employees) also results in estimates of worker effects that are positive, significant, and attenuating with reduced proximity.

[35] Imposing a minimum employment threshold of 1000 for a census tract to be included reduces observations from 5282 to 2696, but the worker results are still positive and mostly significant at the 1% level. Replacing the metropolitan area fixed effects, β_m, with counties, super-PUMAs, and PUMAs increases the number of effects from 12 to 16, 50, and 181 respectively. With the super-PUMAs, the proximity ranges out to 15 minutes still have significant worker effects for the Software, Movie and R&D industries. With the PUMAs, the 0 to 7.5 minute range effects are most robust across industries.

predicted effect of reducing traffic congestion on entry counts. Predictions here find that improved access to qualified workers would substantially increase establishment entries. However, the magnitudes of the increases may be overstated because the present analysis does not account for a variety of feedback effects that might occur if travel times were improved.

These results underscore the role that reducing commute times may play in the prevalence and growth of employment subcenters. Further examination of the importance of nearby workers for intra-urban location choice would benefit from a greater set of control variables (such as the location of input providers), a larger and more diverse set of industries, more urban areas, a greater time span of analysis, and a modeling approach that could incorporate medium and long run feedback effects within urban areas.

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Table 1
Occupational distribution by industry in 2000, for California.

	Industry				
Occupation grouping	Software	Movie	R & D	Instrument	All Others
Management, business and financial operations	0.26	0.19	0.21	0.21	0.14
Computer and mathematical	**0.49***	0.03	0.08	0.08	0.03
Architecture and engineering	0.03	0.01	0.11	**0.21***	0.03
Life, physical, and social science	0.01	0.00	**0.31***	0.02	0.01
Legal	0.00	0.01	0.01	0.00	0.01
Arts, design, entertainment, sports, and media	0.04	**0.33***	0.01	0.02	0.02
Other professional (social serv., edu., health.)	0.01	0.01	0.04	0.00	0.05
Service (health, protect, food, clean, personal)	0.00	0.10	0.02	0.01	0.14
Sales, office, and administrative support	0.13	0.21	0.15	0.17	0.30
Farming, construction, and repair	0.02	0.04	0.02	0.04	0.11
Production	0.01	0.04	0.03	0.23	0.10
Transportation and materials moving	0.00	0.03	0.01	0.01	0.07
Total	1.00	1.00	1.00	1.00	1.00

Notes: Tabulations from Census 2000 5-percent PUMS file for the private sector, for-profit, employed population living in California, age 16 and older. Cells in each industry column give the share of workers in each occupation. Industries are composed of North American Industry Classification System (NAICS) 4-digit industries (Software, 5112 or 5415; Movie, 5121; R&D, 5417; Instrument, 3345). Occupations are groupings of the Standard Occupational Classifications (SOC).
* Indicates the occupation selected to represent qualified workers for each industry.
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Table 2
Aggregations of explanatory and dependent variables for entire study area.

Variable	All	Software	Movie	R & D	Instrument
Employment	13,195,180	180,978	114,568	76,224	104,966
Workers	11,412,949	370,048	343,932	128,941	305,273
New establishment (emp.=1)		567	780	351	89
New establishment (emp.=2 to 499)		5,876	10,526	1,723	394
New establishment (emp.>499)		5	3	1	10
New estab. employ. (emp.=1)		567	780	351	89
New estab. employ. (emp.=2 to 499)		33,626	28,150	11,017	8,324
New estab. employ. (emp.>499)		8,750	10,000	875	10,900
Exiting establishment (emp.=2 to 499)		4,787	2,161	698	459
Exit estab. employ. (emp.=2 to 499)		67,331	20,405	12,091	14,617

Notes: Employment and establishment totals tabulated from National Establishment Time Series (NETS). Worker totals tabulated from Census 2000 Summary File 3, recently working population age 16 and older. Totals computed over 5282 tracts in study area with employment of greater than 100. New establishments are those entering from 2001 to 2004. Exiting establishments are those that stop reporting employment from 2001 to 2004. New establishment count (where employment is 2 to 499) used as dependent variable in subsequent analyses.

Table 3

Summary statistics at census tract level.

Industry	Variable	Mean	Std. Dev.	50th pct.	90th pct.
Software	New establishments (emp.=2 to 499)	1.1	3.4	0.0	3.0
	New estab. employ. (emp.=2 to 499)	6.4	31.7	0.0	10.0
	Same industry emp. (0 to 5 miles)	2.5	4.1	0.9	6.6
	Qualified workers (0 to 5 miles)	5.4	5.5	4.1	10.2
	Same industry emp. (0 to 15 minutes)	2.7	4.8	1.0	6.9
	Qualified workers (0 to 15 minutes)	5.9	6.1	4.5	11.3
Movie	New establishments (emp.=2 to 499)	2.0	10.0	1.0	3.0
	New estab. employ. (emp.=2 to 499)	5.3	30.6	2.0	8.0
	Same industry emp. (0 to 5 miles)	2.9	9.3	0.2	3.5
	Qualified workers (0 to 5 miles)	6.2	7.9	3.5	18.7
	Same industry emp. (0 to 15 minutes)	2.3	7.2	0.2	3.2
	Qualified workers (0 to 15 minutes)	6.0	6.1	4.2	14.9
R & D	New establishments (emp.=2 to 499)	0.3	1.0	0.0	1.0
	New estab. employ. (emp.=2 to 499)	2.1	18.5	0.0	3.0
	Same industry emp. (0 to 5 miles)	0.9	1.4	0.4	2.1
	Qualified workers (0 to 5 miles)	1.8	1.9	1.2	4.6
	Same industry emp. (0 to 15 minutes)	0.9	1.5	0.5	2.2
	Qualified workers (0 to 15 minutes)	1.9	1.7	1.4	4.6
Instrument	New establishments (emp.=2 to 499)	0.1	0.4	0.0	0.0
	New estab. employ. (emp.=2 to 499)	1.6	18.3	0.0	0.0
	Same industry emp. (0 to 5 miles)	1.3	2.5	0.5	3.1
	Qualified workers (0 to 5 miles)	4.1	3.5	3.4	8.0
	Same industry emp. (0 to 15 minutes)	1.5	2.9	0.5	3.6
	Qualified workers (0 to 15 minutes)	4.6	4.3	3.7	8.6
All	Establishments in census tract	214.7	290.8	141.0	424.0
	Same industry emp. (0 to 5 miles)	225.5	193.9	179.4	538.6
	Qualified workers (0 to 5 miles)	176.9	116.1	161.7	327.5
	Same industry emp. (0 to 15 minutes)	220.6	162.6	194.6	455.6
	Qualified workers (0 to 15 minutes)	179.9	102.3	181.3	314.9

Notes: All values calculated at census tract level over 5282 tracts in study area with employment of greater than 100. Statistics give the mean, standard deviation, 50th percentile value, and 90th percentile value for each variable. Establishment and employment statistics calculated with NETS data. Worker statistics calculated with Census 2000 Summary File 3 data. Employment and worker stocks are measured in '000s, for proximity ranges encompassing census tracts in which businesses (for employment data) and residences (for worker data) are located. Great-circle distances used for mile based ranges and vehicle travel time used for minute based ranges.

Table 4

Partial correlations of employment and worker quantities within distance range of 0 to 5 miles.

Quantities within 0 to 5 miles	Same-industry emp.	Other-industry emp.	Qualified workers	Other workers	Same-industry emp.	Other-industry emp.	Qualified workers	Other workers
	Software industry				Movie			
Same-industry employment	1.00				1.00			
Other-industry employment	0.46	1.00			0.35	1.00		
Qualified workers	0.65	0.61	1.00		0.81	0.67	1.00	
Other workers	0.21	0.87	0.60	1.00	0.31	0.86	0.63	1.00
	R & D industry				Instrument industry			
Same-industry employment	1.00				1.00			
Other-industry employment	0.49	1.00			0.21	1.00		
Qualified workers	0.61	0.57	1.00		0.50	0.49	1.00	
Other workers	0.35	0.87	0.53	1.00	0.05	0.87	0.55	1.00

Notes: Partial correlations are conditional on metropolitan areas. Employment stocks calculated with NETS data. Worker stocks calculated with Census 2000 Summary File 3 data. All values calculated at census tract level over 5282 tracts in study area with employment of greater than 100.

Table 5
The effect of employment and workers on establishment entry. Proximity measured with distance.

	Software		Movie		R & D		Instrument	
Covariate	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Log of establishment count								
Within tract	1.010***	0.996***	0.840***	0.771***	1.076***	1.014***	1.070***	1.063***
	(0.026)	(0.026)	(0.034)	(0.028)	(0.044)	(0.041)	(0.071)	(0.072)
Same-industry employment in								
0 to 1 mile	0.291***	0.248***	0.096***	0.016	0.298***	0.216***	0.390***	0.294***
	(0.066)	(0.069)	(0.030)	(0.014)	(0.083)	(0.061)	(0.086)	(0.103)
1 to 2.5 miles	0.063***	0.002	0.049***	-0.027***	0.191***	0.067	0.183***	0.083
	(0.023)	(0.023)	(0.007)	(0.009)	(0.072)	(0.082)	(0.047)	(0.064)
2.5 to 5 miles	0.063***	0.006	0.044***	-0.026***	0.156***	0.039*	0.073**	0.012
	(0.012)	(0.016)	(0.006)	(0.006)	(0.020)	(0.023)	(0.032)	(0.043)
5 to 10 miles	0.032***	0.007	0.017***	-0.011***	0.046**	-0.016	0.046*	0.044
	(0.007)	(0.010)	(0.004)	(0.004)	(0.022)	(0.021)	(0.026)	(0.031)
Other-industry employment in								
0 to 1 mile	-0.005***	-0.003*	-0.002**	-0.002**	-0.002**	-0.001	-0.002	-0.001
	(0.002)	(0.002)	(0.001)	(0.001)	(0.001)	(0.001)	(0.002)	(0.002)
1 to 2.5 miles	-0.000	0.002**	-0.000	-0.000	-0.001*	0.001	-0.002	-0.000
	(0.001)	(0.001)	(0.001)	(0.000)	(0.001)	(0.001)	(0.002)	(0.002)
2.5 to 5 miles	-0.001***	0.001*	-0.001*	0.000	-0.001	0.001**	-0.003***	-0.001
	(0.000)	(0.001)	(0.000)	(0.000)	(0.000)	(0.001)	(0.001)	(0.001)
5 to 10 miles	-0.000*	0.001**	-0.001***	0.000	0.000	0.001	-0.000	-0.001
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.001)
Qualified workers in								
0 to 2.5 miles		0.124***		0.211***		0.241***		-0.019
		(0.037)		(0.019)		(0.053)		(0.101)
2.5 to 5 miles		0.016		0.069***		0.136***		0.134**
		(0.018)		(0.012)		(0.049)		(0.067)
5 to 10 miles		0.013		0.043***		0.051*		-0.012
		(0.009)		(0.006)		(0.026)		(0.026)
Other workers in								
0 to 2.5 miles		-0.011***		-0.014***		-0.009***		-0.003
		(0.003)		(0.002)		(0.002)		(0.005)
2.5 to 5 miles		-0.003**		-0.002**		-0.004***		-0.009**
		(0.001)		(0.001)		(0.001)		(0.004)
5 to 10 miles		-0.001**		-0.001***		-0.000		0.002*
		(0.001)		(0.000)		(0.001)		(0.001)

Notes: Coefficient estimates for negative binomial estimation with new establishment count as dependent variable. Robust-clustered standard errors (in parentheses) clustered in 181 groups at the 5-percent PUMA level. Choice set includes 5282 census tracts in study area with employment of greater than 100. Metropolitan area fixed effects and a constant term are not reported here. Odd columns estimate Eq. (5) with only employment proximity ranges, while even columns estimate Eq. (6) including both employment and worker proximity ranges. Great-circle distances used for proximity ranges.
* Significant at the 10% level. ** Significant at the 5% level. *** Significant at the 1% level.

Table 6

The effect of employment and workers on new establishment entry. Proximity measured by travel time.

	Software		Movie		R & D		Instrument	
Covariate	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Log of establishment count								
Within tract	1.008***	0.982***	0.837***	0.745***	1.060***	0.994***	1.071***	1.063***
	(0.026)	(0.026)	(0.032)	(0.028)	(0.043)	(0.038)	(0.070)	(0.072)
Same-industry employment in								
0 to 1 mile	0.272***	0.285***	0.138***	0.039*	0.290***	0.194***	0.408***	0.376***
	(0.065)	(0.064)	(0.049)	(0.022)	(0.079)	(0.044)	(0.082)	(0.088)
out to 7.5 min.	0.176***	0.054	0.068***	-0.011*	0.124**	-0.058*	0.217***	0.192***
	(0.045)	(0.035)	(0.009)	(0.006)	(0.060)	(0.035)	(0.074)	(0.073)
7.5 to 15 min.	0.052***	-0.012	0.044***	-0.024***	0.132***	0.022	0.131***	0.107***
	(0.010)	(0.011)	(0.010)	(0.007)	(0.026)	(0.029)	(0.029)	(0.034)
15 to 30 min.	0.021***	0.002	0.022***	-0.011***	0.021	-0.013	0.044***	0.053**
	(0.005)	(0.008)	(0.004)	(0.003)	(0.015)	(0.015)	(0.013)	(0.024)
Other-industry employment in								
0 to 1 mile	-0.003*	-0.004**	-0.001***	-0.001*	-0.001	-0.000	-0.001	-0.001
	(0.002)	(0.002)	(0.000)	(0.000)	(0.001)	(0.001)	(0.002)	(0.002)
out to 7.5 min.	-0.003**	0.001	-0.004***	-0.000	-0.003*	-0.001	-0.006*	-0.004
	(0.001)	(0.001)	(0.001)	(0.001)	(0.002)	(0.001)	(0.003)	(0.003)
7.5 to 15 min.	-0.001***	0.001*	-0.000	0.000	-0.001	0.001*	-0.003***	-0.002**
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.001)	(0.001)
15 to 30 min.	-0.000	0.001**	-0.001***	0.000**	0.000	0.000**	-0.000	-0.001*
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Qualified workers in								
0 to 7.5 min.		0.191***		0.232***		0.247***		-0.040
		(0.044)		(0.025)		(0.058)		(0.114)
7.5 to 15 min.		0.037***		0.066***		0.190***		0.031
		(0.011)		(0.014)		(0.038)		(0.043)
15 to 30 min.		0.015**		0.038***		0.036*		-0.005
		(0.007)		(0.006)		(0.020)		(0.020)
Other workers in								
0 to 7.5 min.		-0.013***		-0.018***		-0.010***		-0.001
		(0.002)		(0.002)		(0.002)		(0.005)
7.5 to 15 min.		-0.004***		-0.002***		-0.005***		-0.003
		(0.001)		(0.001)		(0.001)		(0.002)
15 to 30 min.		-0.001**		-0.002***		0.000		0.002**
		(0.000)		(0.000)		(0.000)		(0.001)

Notes: Coefficient estimates for negative binomial estimation with new establishment count as dependent variable. Robust-clustered standard errors (in parentheses) clustered in 181 groups at the 5-percent PUMA level. Choice set includes 5282 census tracts in study area with employment of greater than 100. Metropolitan area fixed effects and a constant term are not reported here. Odd columns estimate Eq. (5) with only employment proximity ranges, while even columns estimate Eq. (6) including both employment and worker proximity ranges. Distance in miles or travel time in minutes used for proximity ranges as noted.

* Significant at the 10% level. ** Significant at the 5% level. *** Significant at the 1% level.

Table 7
Predicted effects on entry of increase from 50th to 90th percentile in same-industry employment or qualified workers. Proximity measured by travel time.

	Software		Movie		R & D		Instrument	
Covariate	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Log of establishment count								
Within tract	5.690***	5.545***	4.724***	4.205***	5.984***	5.614***	6.049***	6.002***
	(0.146)	(0.146)	(0.183)	(0.160)	(0.243)	(0.215)	(0.394)	(0.407)
Same-industry employment in								
0 to 1 mile	0.066***	0.069***	0.011***	0.003*	0.022***	0.015***	0.028***	0.026***
	(0.016)	(0.015)	(0.004)	(0.002)	(0.006)	(0.003)	(0.006)	(0.006)
out to 7.5 min.	0.107***	0.033	0.021***	-0.003*	0.030**	-0.014*	0.102***	0.091***
	(0.028)	(0.021)	(0.003)	(0.002)	(0.015)	(0.008)	(0.035)	(0.034)
7.5 to 15 min.	0.265***	-0.062	0.130***	-0.070***	0.230***	0.038	0.340***	0.278***
	(0.051)	(0.057)	(0.028)	(0.020)	(0.045)	(0.050)	(0.075)	(0.090)
15 to 30 min.	0.294***	0.034	0.691***	-0.339***	0.126	-0.076	0.266***	0.319**
	(0.067)	(0.110)	(0.120)	(0.104)	(0.086)	(0.090)	(0.076)	(0.143)
Other-industry employment in								
0 to 1 mile	-0.057*	0.034**	0.691***	-0.339*	0.126	-0.076	0.266	0.319
	(0.031)	(0.110)	(0.120)	(0.104)	(0.086)	(0.090)	(0.076)	(0.143)
out to 7.5 min.	-0.163**	-0.079	-0.027***	-0.017	-0.021*	-0.004	-0.029*	-0.027
	(0.076)	(0.031)	(0.009)	(0.009)	(0.022)	(0.014)	(0.030)	(0.034)
7.5 to 15 min.	-0.261***	0.034*	-0.185	-0.006	-0.144	-0.065*	-0.304***	-0.216**
	(0.091)	(0.068)	(0.068)	(0.037)	(0.082)	(0.075)	(0.174)	(0.168)
15 to 30 min.	-0.112	0.178**	-0.056***	0.042**	-0.148	0.182**	-0.658	-0.510*
	(0.088)	(0.102)	(0.070)	(0.067)	(0.091)	(0.101)	(0.174)	(0.228)
Qualified workers in								
0 to 7.5 min.		0.308***		0.524***		0.141***		-0.050
		(0.071)		(0.055)		(0.033)		(0.142)
7.5 to 15 min.		0.223***		0.570***		0.502***		0.117
		(0.069)		(0.119)		(0.099)		(0.160)
15 to 30 min.		0.302**		1.038***		0.208*		-0.061
		(0.136)		(0.157)		(0.114)		(0.217)
Other workers in								
0 to 7.5 min.		-0.590***		-0.772***		-0.439***		-0.050
		(0.107)		(0.079)		(0.096)		(0.247)
7.5 to 15 min.		-0.417***		-0.197***		-0.496***		-0.343
		(0.102)		(0.075)		(0.092)		(0.261)
15 to 30 min.		-0.274**		-0.640***		0.005		0.583**
		(0.124)		(0.126)		(0.127)		(0.283)

Notes: Predicted effect on entry is for increase from the 50th to 90th percentile in each proximity range using coefficient from Table 6. Standard errors of predicted effects in parentheses. Each predicted effect assumes all other variables held constant.
* Significant at the 10% level. ** Significant at the 5% level. *** Significant at the 1% level.

Table 8

Aggregate observed and predicted entry under congested and uncongested travel time scenarios.

Scenario for new establishment entry count	Software	Movie	R & D	Instrument
Full study area				
Observed	5,876	10,526	1,723	394
Predicted, peak travel times	5,945	10,003	1,712	393
Predicted, off peak times	6,070	10,796	1,900	400
Predicted, off peak times, same industry	5,953	8,928	1,707	448
Predicted, off peak times, other industry	6,736	11,348	1,960	316
Predicted, off peak times, qualified workers	6,876	19,802	2,042	396
Predicted, off peak times, other workers	4,724	6,165	1,429	444
Selected district for each industry				
Observed	286	240	66	7.0
Predicted, peak travel times	305	238	65	4.2
Predicted, off peak times	341	293	97	3.9
Predicted, off peak times, same industry	295	216	65	4.5
Predicted, off peak times, other industry	328	296	75	3.0
Predicted, off peak times, qualified workers	394	458	117	4.1
Predicted, off peak times, other workers	255	139	48	4.9

Notes: New establishment entry counts are for three scenarios. First, observed entry counts are reported. Second, predicted counts are reported with coefficients from Table 6 and covariates based on peak-period travel times. Third, predicted counts are reported using coefficients estimated on peak-period travel times (still from Table 6) and covariates based on off-peak period travel times. For the third scenario, predicted counts are also reported for cases where only one set of covariates, in turn, is set to off-peak period travel times. Full study area includes 5282 census tracts with employment of greater than 100. Selected districts are an individual 5-percent PUMA for each industry. The districts include (PUMA): Mountain View (0602701) in Santa Clara County, for the Software industry; Burbank (0604800) in Los Angeles County, for the Movie industry; La Jolla (0608110) in San Diego County, for the R&D industry; and Irvine (0605420) in Orange County, for the Instrument industry.

Table 9
Alternative specifications: Entry limited to 2001 or 2002.

	Software		Movie		R & D		Instrument	
Covariate	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Log of establishment count								
Within tract	0.995***	0.961***	0.872***	0.766***	1.085***	0.996***	1.060***	1.042***
	(0.029)	(0.029)	(0.036)	(0.034)	(0.065)	(0.060)	(0.083)	(0.084)
Same-industry employment in								
0 to 1 mile	0.247***	0.256***	0.137***	0.031	0.369***	0.257***	0.397***	0.346***
	(0.066)	(0.064)	(0.051)	(0.024)	(0.091)	(0.065)	(0.069)	(0.086)
out to 7.5 min.	0.170***	0.041	0.070***	-0.017**	0.081	-0.078	0.217***	0.192**
	(0.043)	(0.034)	(0.011)	(0.007)	(0.093)	(0.072)	(0.072)	(0.076)
7.5 to 15 min.	0.053***	-0.013	0.047***	-0.028***	0.177***	0.093***	0.175***	0.141***
	(0.010)	(0.011)	(0.011)	(0.008)	(0.035)	(0.030)	(0.035)	(0.044)
15 to 30 min.	0.020***	0.002	0.024***	-0.012***	0.031	0.008	0.048***	0.052
	(0.005)	(0.008)	(0.004)	(0.004)	(0.019)	(0.020)	(0.018)	(0.033)
Other-industry employment in								
0 to 1 mile	-0.003*	-0.004**	-0.001**	-0.001	-0.001	-0.000	-0.001	-0.000
	(0.002)	(0.002)	(0.001)	(0.001)	(0.001)	(0.001)	(0.002)	(0.002)
out to 7.5 min.	-0.004**	0.001	-0.004***	0.000	-0.003	0.001	-0.003	-0.001
	(0.001)	(0.001)	(0.001)	(0.001)	(0.002)	(0.002)	(0.004)	(0.003)
7.5 to 15 min.	-0.001***	0.001*	-0.000	0.000	-0.001**	0.001	-0.004***	-0.003**
	(0.000)	(0.000)	(0.000)	(0.000)	(0.001)	(0.001)	(0.001)	(0.001)
15 to 30 min.	-0.000	0.001**	-0.001***	0.000**	0.000	0.000	-0.000	-0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.001)
Qualified workers in								
0 to 7.5 min.		0.212***		0.240***		0.128		-0.069
		(0.046)		(0.027)		(0.099)		(0.119)
7.5 to 15 min.		0.032***		0.075***		0.161***		0.040
		(0.012)		(0.016)		(0.061)		(0.049)
15 to 30 min.		0.014**		0.042***		0.000		-0.001
		(0.007)		(0.007)		(0.028)		(0.024)
Other workers in								
0 to 7.5 min.		-0.016***		-0.019***		-0.015***		-0.001
		(0.002)		(0.002)		(0.003)		(0.007)
7.5 to 15 min.		-0.004***		-0.003***		-0.005***		-0.005
		(0.001)		(0.001)		(0.002)		(0.003)
15 to 30 min.		-0.001**		-0.002***		0.001		0.001
		(0.000)		(0.000)		(0.001)		(0.001)

Notes: Coefficient estimates are for negative binomial estimation with new establishment count in 2001 and 2002 only as dependent variable (as opposed to 2001 through 2004 in earlier specifications).
* Significant at the 10% level. ** Significant at the 5% level. *** Significant at the 1% level.

Table 10
Alternative specifications: New establishment employment.

	Software		Movie		R & D		Instrument	
Covariate	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Log of establishment count								
Within tract	1.077***	1.053***	0.807***	0.733***	1.165***	1.141***	1.370***	1.400***
	(0.065)	(0.063)	(0.039)	(0.036)	(0.066)	(0.064)	(0.118)	(0.121)
Same-industry employment in								
0 to 1 mile	0.439***	0.480***	0.192***	0.094***	1.249*	0.769	3.840***	2.803**
	(0.109)	(0.105)	(0.048)	(0.019)	(0.755)	(0.584)	(1.306)	(1.132)
out to 7.5 min.	0.312***	0.110	0.083***	0.010	0.356*	-0.011	0.151	0.210
	(0.090)	(0.090)	(0.013)	(0.010)	(0.197)	(0.095)	(0.162)	(0.186)
7.5 to 15 min.	0.084***	-0.018	0.042***	-0.022**	0.142***	0.003	0.192**	0.221**
	(0.018)	(0.024)	(0.011)	(0.009)	(0.044)	(0.048)	(0.077)	(0.086)
15 to 30 min.	0.033**	-0.001	0.024***	-0.004	0.049*	0.022	0.041	0.120**
	(0.015)	(0.021)	(0.004)	(0.006)	(0.025)	(0.026)	(0.034)	(0.057)
Other-industry employment in								
0 to 1 mile	-0.001	-0.003	0.001	0.003	-0.003	-0.002	-0.003	-0.003
	(0.002)	(0.002)	(0.002)	(0.002)	(0.003)	(0.003)	(0.003)	(0.003)
out to 7.5 min.	-0.004*	0.003	-0.003***	-0.000	-0.004**	-0.001	-0.000	-0.003
	(0.002)	(0.002)	(0.001)	(0.001)	(0.002)	(0.002)	(0.005)	(0.006)
7.5 to 15 min.	-0.002***	0.001	-0.000	-0.001	-0.001	0.001	-0.005***	-0.007***
	(0.001)	(0.001)	(0.000)	(0.000)	(0.001)	(0.001)	(0.002)	(0.002)
15 to 30 min.	-0.001*	0.000	-0.001***	0.000	0.000	0.001***	-0.001**	-0.002**
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.001)	(0.001)
Qualified workers in								
0 to 7.5 min.		0.295***		0.249***		0.519***		-0.668***
		(0.087)		(0.036)		(0.174)		(0.255)
7.5 to 15 min.		0.061**		0.052**		0.156**		0.221*
		(0.024)		(0.021)		(0.068)		(0.126)
15 to 30 min.		0.030*		0.031***		-0.029		-0.098**
		(0.017)		(0.007)		(0.032)		(0.047)
Other workers in								
0 to 7.5 min.		-0.023***		-0.016***		-0.013***		0.016
		(0.004)		(0.002)		(0.004)		(0.011)
7.5 to 15 min.		-0.005***		-0.000		-0.004**		-0.005
		(0.002)		(0.001)		(0.002)		(0.004)
15 to 30 min.		-0.001		-0.001***		-0.000		0.004**
		(0.001)		(0.000)		(0.001)		(0.002)

Notes: Coefficient estimates are for negative binomial estimation with new establishment employment as dependent variable (as opposed to new establishment count in earlier specifications).
* Significant at the 10% level. ** Significant at the 5% level. *** Significant at the 1% level.

Table 11
Alternative specifications: Normalized for census tract land area.

	Software		Movie		R & D		Instrument	
Covariate	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Log of establishment count divided by area								
Within tract	0.003***	0.003***	0.003***	0.002***	0.002***	0.002***	0.002***	0.002***
	(0.001)	(0.001)	(0.001)	(0.000)	(0.001)	(0.000)	(0.000)	(0.000)
Same-industry employment in								
0 to 1 mile	0.375***	0.433***	0.129**	0.027	0.371***	0.267**	0.773**	0.884**
	(0.120)	(0.115)	(0.054)	(0.018)	(0.136)	(0.110)	(0.306)	(0.347)
out to 7.5 min.	0.288***	0.136**	0.076***	-0.034***	0.225*	-0.040	0.206***	0.234**
	(0.067)	(0.055)	(0.013)	(0.012)	(0.121)	(0.059)	(0.071)	(0.092)
7.5 to 15 min.	0.064***	0.012	0.052***	-0.056***	0.155***	0.037	0.117***	0.167***
	(0.013)	(0.018)	(0.011)	(0.010)	(0.038)	(0.036)	(0.033)	(0.041)
15 to 30 min.	0.025***	0.012	0.019***	-0.028***	0.057***	0.024	0.068***	0.127***
	(0.008)	(0.012)	(0.004)	(0.005)	(0.022)	(0.024)	(0.018)	(0.033)
Other-industry employment in								
0 to 1 mile	-0.005**	-0.006***	-0.003*	-0.001	0.001	0.002	-0.003	-0.001
	(0.002)	(0.002)	(0.002)	(0.002)	(0.003)	(0.002)	(0.004)	(0.004)
out to 7.5 min.	0.002	0.003*	0.002	0.003***	0.004**	0.003	0.006*	0.003
	(0.001)	(0.002)	(0.002)	(0.001)	(0.002)	(0.002)	(0.003)	(0.003)
7.5 to 15 min.	0.000	0.000	0.002***	0.000	0.001*	0.000	-0.001	-0.003**
	(0.000)	(0.001)	(0.000)	(0.000)	(0.000)	(0.001)	(0.001)	(0.001)
15 to 30 min.	0.000	0.000	-0.000**	0.000*	0.000	-0.000	0.000	-0.001***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.001)
Qualified workers in								
0 to 7.5 min.		0.333***		0.377***		0.542***		0.063
		(0.082)		(0.045)		(0.095)		(0.144)
7.5 to 15 min.		0.034*		0.094***		0.273***		-0.006
		(0.018)		(0.019)		(0.051)		(0.062)
15 to 30 min.		0.015		0.056***		0.085***		-0.052**
		(0.011)		(0.007)		(0.030)		(0.025)
Other workers in								
0 to 7.5 min.		-0.004		-0.006**		0.002		0.012**
		(0.003)		(0.002)		(0.003)		(0.006)
7.5 to 15 min.		-0.001		-0.001		-0.003**		0.001
		(0.001)		(0.001)		(0.001)		(0.003)
15 to 30 min.		0.000		-0.001***		0.001*		0.003***
		(0.001)		(0.000)		(0.000)		(0.001)

Notes: Coefficient are for negative binomial estimation estimated effects on new establishment count normalized by census tract land area (km sq.) as an exposure variable. Within tract establishment count divided by land area Earlier specifications used the absolute count of new establishments.
* Significant at the 10% level. ** Significant at the 5% level. *** Significant at the 1% level.

Table 12
Alternative specifications: Zip Code based analysis.

	Software		Movie		R & D		Instrument	
Covariate	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Log of establishment count								
Within tract	1.042***	1.057***	1.019***	0.965***	1.031***	0.982***	1.057***	1.053***
	(0.058)	(0.057)	(0.055)	(0.054)	(0.074)	(0.068)	(0.116)	(0.117)
Same-industry employment in								
0 to 1 mile	0.039***	0.033***	0.017**	0.009**	0.010	0.004	0.039***	0.037***
	(0.010)	(0.009)	(0.007)	(0.004)	(0.013)	(0.005)	(0.013)	(0.013)
out to 7.5 min.	0.024**	0.009	0.007**	-0.001	0.082	0.031	-0.037	-0.031
	(0.011)	(0.010)	(0.003)	(0.002)	(0.053)	(0.043)	(0.053)	(0.053)
7.5 to 15 min.	0.004***	-0.001	0.006***	0.000	0.009	-0.008**	0.008***	0.008
	(0.001)	(0.002)	(0.001)	(0.001)	(0.007)	(0.004)	(0.003)	(0.006)
15 to 30 min.	0.001	0.003**	0.002***	-0.001	-0.000	-0.004**	0.002	0.005*
	(0.001)	(0.001)	(0.000)	(0.000)	(0.002)	(0.002)	(0.003)	(0.003)
Other-industry employment in								
0 to 1 mile	-0.001**	-0.000**	-0.000**	-0.000**	0.000	0.000	-0.001	-0.001
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
out to 7.5 min.	0.000	0.000	-0.001***	-0.000***	-0.000	-0.000**	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
7.5 to 15 min.	-0.000***	0.000	0.000	0.000	-0.000	0.000***	-0.000***	-0.000*
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
15 to 30 min.	-0.000*	-0.000	-0.000***	0.000*	-0.000	0.000***	0.000	-0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Qualified workers in								
0 to 7.5 min.		0.022**		0.030***		0.033***		0.014
		(0.009)		(0.005)		(0.013)		(0.017)
7.5 to 15 min.		0.002		0.003*		0.021***		-0.002
		(0.002)		(0.002)		(0.005)		(0.006)
15 to 30 min.		-0.002**		0.003***		0.002		-0.003*
		(0.001)		(0.001)		(0.002)		(0.002)
Other workers in								
0 to 7.5 min.		-0.001**		-0.001***		-0.001*		-0.000
		(0.000)		(0.000)		(0.000)		(0.001)
7.5 to 15 min.		-0.000***		-0.000		-0.001***		-0.000
		(0.000)		(0.000)		(0.000)		(0.000)
15 to 30 min.		0.000		-0.000***		-0.000**		0.000***
		(0.000)		(0.000)		(0.000)		(0.000)

Notes: Coefficient estimates are for negative binomial estimation with new establishment location choice at the level of Zip Codes, rather than census tracts. Choice set includes 824 Zip Codes in study area with employment of greater than 100.
* Significant at the 10% level. ** Significant at the 5% level. *** Significant at the 1% level.